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הפקולטה למדעי ההנדסה
המחלקה להנדסת מערכות תוכנה ומידע

אוניברסיטת בן-גוריון בנגב
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Explainability In Medical Prediction Models

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Project Steps

1 Literature Review
Review of explainability techniques and definition of research questions

2 Model Training
Training XGBoost, Logistic Regression, and Decision Tree on GOSSIS data

3 Explanation Generation
Applying SHAP, LIME, and inherent methods to each model.

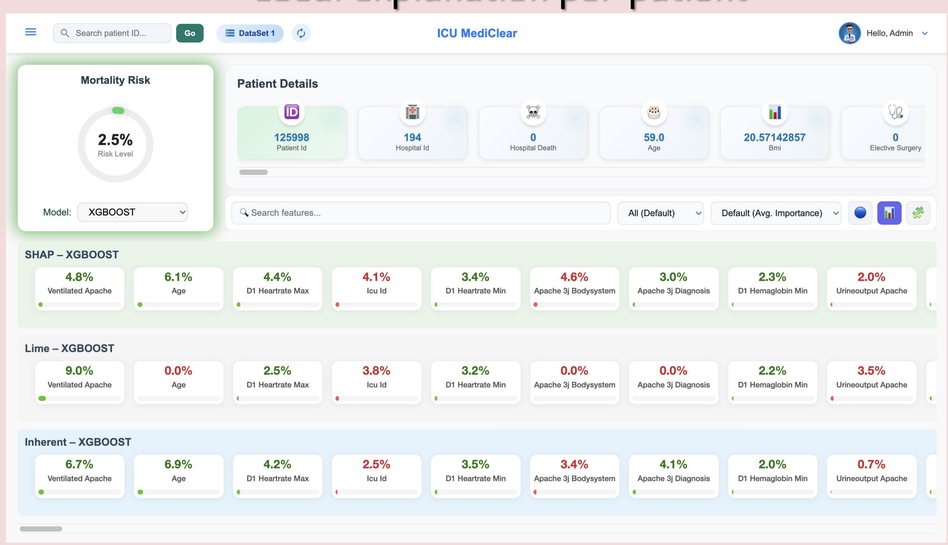
4 Dashboard Development
Creating an interactive tool for comparing explanations

5 Correlation Analysis
Evaluating consistency using Kendall's W, Pearson, Sign Agreement, and Intersection@K

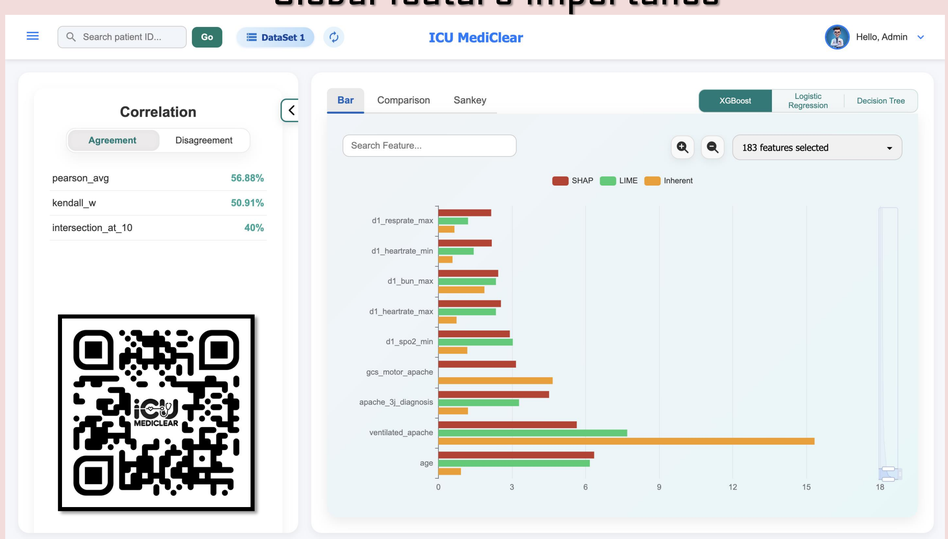
6 Conclusions
Creating an interactive tool for comparing explanations

Interactive Dashboard

Local explanation per patient



Global feature importance



Background & Motivation

AI models are increasingly used in clinical decision-making, but their complexity often creates a **"black box"** problem. To build trust, explainability techniques (SHAP, LIME, inherent methods) must provide consistent, clear insights. This research focuses on evaluating their consistency and agreement for **ICU** mortality prediction using the **"GOSSIS"** and **"Diabetes 130-US Hospitals for Years 1999-2008"** datasets

Research Objective

- Evaluate how consistent different explanation methods are, globally and locally.
- Identify when methods agree/disagree on key features.
- Build a dashboard for clinical and research use, enabling intuitive comparison.

Correlation Metrics

Metric	Purpose
Kendall's W	Agreement on feature ranking
Pearson Correlation	Similarity of feature importance values
Intersection@K	Top-K feature overlap across methods (K=10)
Sign Agreement	(Local only) Whether methods agree on direction (+/–)

Results

Global Consistency

Model	Kendall's W	Intersection@10	Pearson Avg
Logistic Regression	0.96	0.9	0.97
Decision Tree	0.95	0.90	0.96
XGBoost	0.83	0.8	0.88

Local Consistency- Average

Model	Kendall's W	Intersection@10	Pearson Avg	Sign Agreement
Logistic Regression	0.8875	0.5794	0.6657	0.9863
Decision Tree	0.9002	0.5025	0.7878	0.9709
XGBoost	0.8999	0.5638	0.6943	0.9752

Methodology

Models Used
- Logistic Regression (interpretable)
- Decision Tree (interpretable)
- XGBoost (black-box)

Explanation Methods
- SHAP
- LIME
- Inherent (native logic of each model)

Normalization
- All feature contributions normalized to 100% per method for fair comparison.

Conclusion

1. **At a global level,** explanation methods show higher agreement on simpler models (such as Logistic Regression and Decision Trees) than on complex models like XGBoost.
2. There is **high correlation** between explanation methods.
3. **Interactive tools** can help clinicians better interpret AI predictions and trust the model outputs.